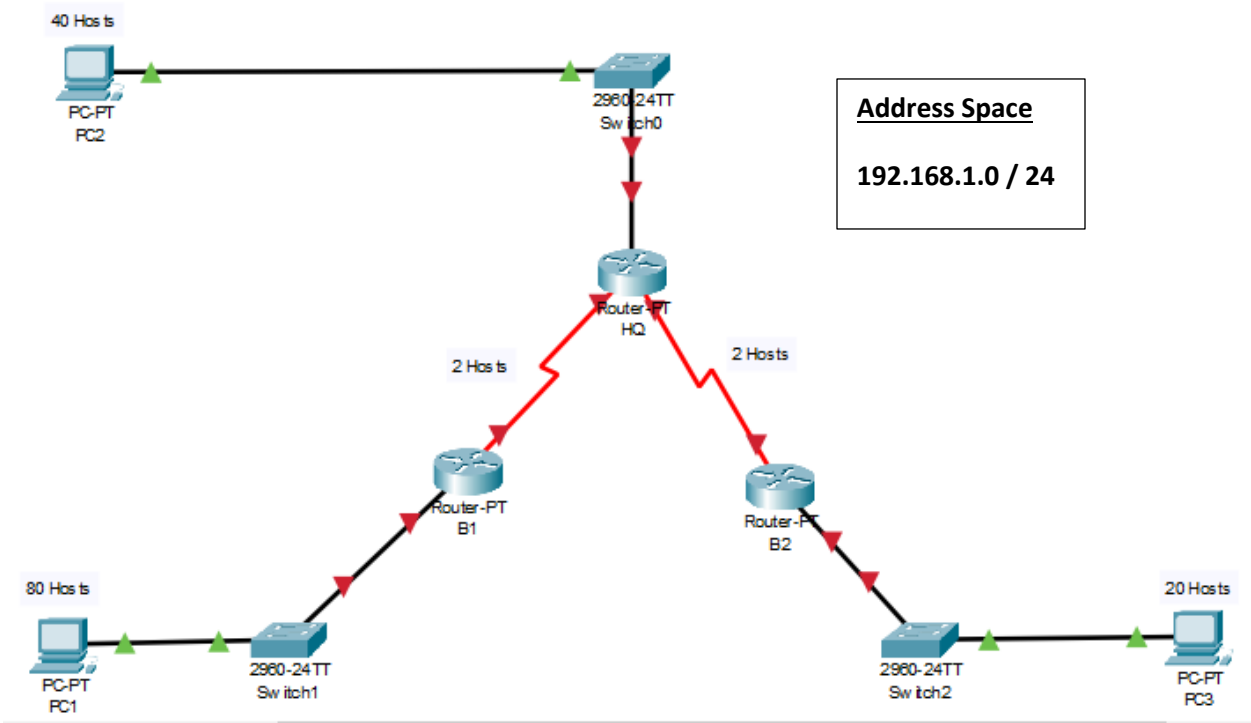
## Implementation of different types of LAN topology with IP addressing using PT

### Objectives:

- Design the following topology in Packet Tracer.
- Subnet each of the network with the IP and complete the following table.
- Assign the appropriate IP into each of the network.
- Add one more PC to each of router and assign IP.
- Test connectivity between PC in a router.




First subnet (80 hosts)

192.168.1.0 /24

Required hosts 80= $2^7$ =128 hosts

Subnet mask= 11111111. 11111111.11111111. 10000000 =255.255.255.128

N.A= 192.168.1.0 /25 ------- B.A= 192.168.1.127 /25

Second subnet (40 hosts)

192.168.1.128 /26

Required hosts 40= $2^6$ =64 hosts

Subnet mask= 11111111. 11111111.11111111. 11000000 =255.255.255.192

N.A= 192.168.1.128 /26 -------- B.A= 192.168.1.191 /26

3rd subnet (20 hosts)

192.168.1.192 /27

Required hosts 20= $2^5$ =32 hosts

Subnet mask= 11111111. 11111111.11111111. 11100000 =255.255.255.224

N.A= 192.168.1.192 /27 ---------- B.A= 192.168.1.223 /27

$4^{th}$ subnet (2 hosts)

192.168.1.224 /30

Required hosts 20= $2^2$ =4 hosts

Subnet mask= 11111111. 11111111.11111111. 11111100 =255.255.255.252

N.A= 192.168.1.224 /30 ---------- B.A= 192.168.1.227/30

$5^{th}$ subnet (2 hosts)

192.168.1.228 /30

Required hosts 20= $2^2$ =4 hosts

Subnet mask= 11111111. 11111111.11111111. 11111100 =255.255.255.252

N.A= 192.168.1.228 /30 ---------- B.A= 192.168.1.231/30

Subnet Table

| Subnet | Assignable Adress Range | Subnet Mask | Subnet Address | Broadcast Address |
|---|---|---|---|---|
| First subnet (80 hosts) | 192.168.1.1 /25 - 192.168.1.126 /25 | 255.255.255.128 | 192.168.1.0 /25 | 192.168.1.127 /25 |
| Second subnet (40 hosts) | 192.168.1.129 /26 - 192.168.1.190 /26 | 255.255.255.192 | 192.168.1.128 /26 | 192.168.1.191 /26 |
| 3rd subnet (20 hosts) | 192.168.1.193 /27- 192.168.1.222 /27 | 255.255.255.224 | 192.168.1.192 /27 | 192.168.1.223 /27 |
| 4th subnet (2 hosts) | 192.168.1.225 /30- 192.168.1.226/30 | 255.255.255.252 | 192.168.1.224 /30 | 192.168.1.227/30 |
| 5th subnet (2 hosts) | 192.168.1.229/30- 192.168.1.230/30 | 255.255.255.252 | 192.168.1.228 /30 | 192.168.1.231 /30 |

Addressing table

| Device | Interface | IP address | Subnet mask | Default Gateway |
|---|---|---|---|---|
| HQ | Fast ethernet 0/0 | 192.168.1.190 /26 | 255.255.255.192 | N/A |
| | Serial 2/0 | 192.168.1.225 /30 | 255.255.255.252 | |
| | Serial 3/0 | 192.168.1.229/30 | 255.255.255.252 | |
| B1 | Fast ethernet 0/0 | 192.168.1.126 /25 | 255.255.255.128 | |
| | Serial 2/0 | 192.168.1.226/30 | 255.255.255.252 | |
| B2 | Fast ethernet 0/0 | 192.168.1.222 /27 | 255.255.255.224 | |
| | Serial 3/0 | 192.168.1.230/30 | 255.255.255.252 | |
| PC1 | N/A | 192.168.1.1 /25 | 255.255.255.128 | 192.168.1.126 /25 |
| PC2 | N/A | 192.168.1.129 /26 | 255.255.255.192 | 192.168.1.190 /26 |
| PC3 | N/A | 192.168.1.193/27 | 255.255.255.224 | 192.168.1.222 /27 |

**Static Routing Table for Router**

| Router | Unknown Network Address | Mask Address | Next Hop |
|---|---|---|---|
| **HQ** | 192.168.1.0 /25 | 255.255.255.128 | 192.168.1.226/30 |
| | 192.168.1.192/27 | 255.255.255.224 | 192.168.1.230/30 |
| **B1** | 192.168.1.128 /26 | 255.255.255.192 | 192.168.1.225 /30 |
| | 192.168.1.192 /27 | **255.255.255.224** | 192.168.1.225 /30 |
| | 192.168.1.228 /30 | **255.255.255.252** | 192.168.1.225 /30 |
| B2 | 192.168.1.128 /26 | 255.255.255.192 | 192.168.1.229/30 |
| | 192.168.1.0 /25 | 255.255.255.128 | 192.168.1.229/30 |
| | 192.168.1.224 /30 | **255.255.255.252** | 192.168.1.229/30 |